An adaptive, uncertainty-aware framework for regional seismic risk assessment integrating microzonation and multi-level vulnerability and exposure modeling

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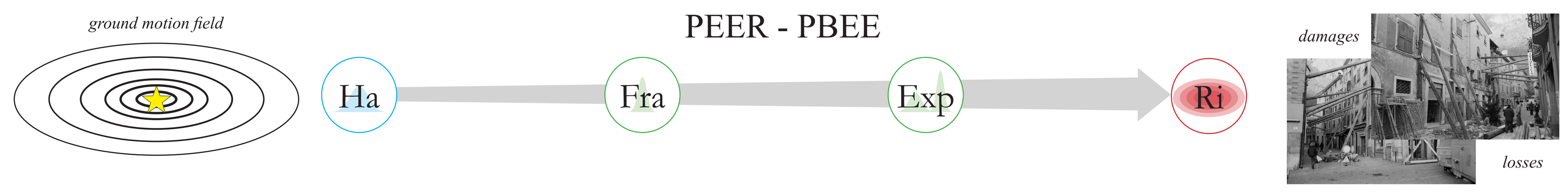
CONTEXT AND MOTIVATIONS

Ha How much do local effects matters in the regional risk assessment of a sedimetary valley basin?

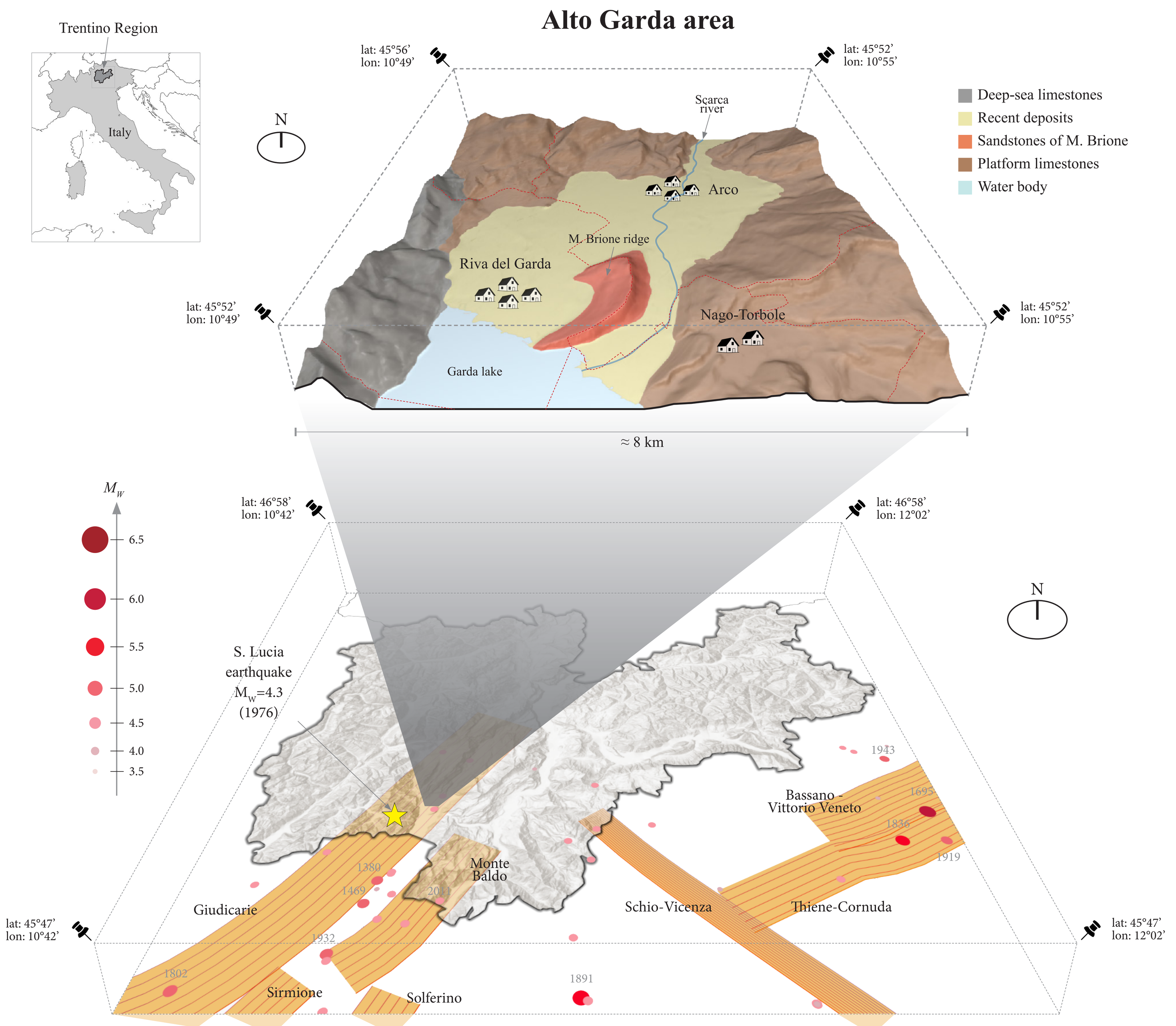
Fra How do we account for uncertainties in fragility models? How should model selection be performed?

Exp How do we account for uncertainties in exposure models? What is the impact of performing a local refinement?

We tried to give an asnswer by reproducing the historical "Santa Lucia" earthquake, occurred in 1976 in the Alto Garda region

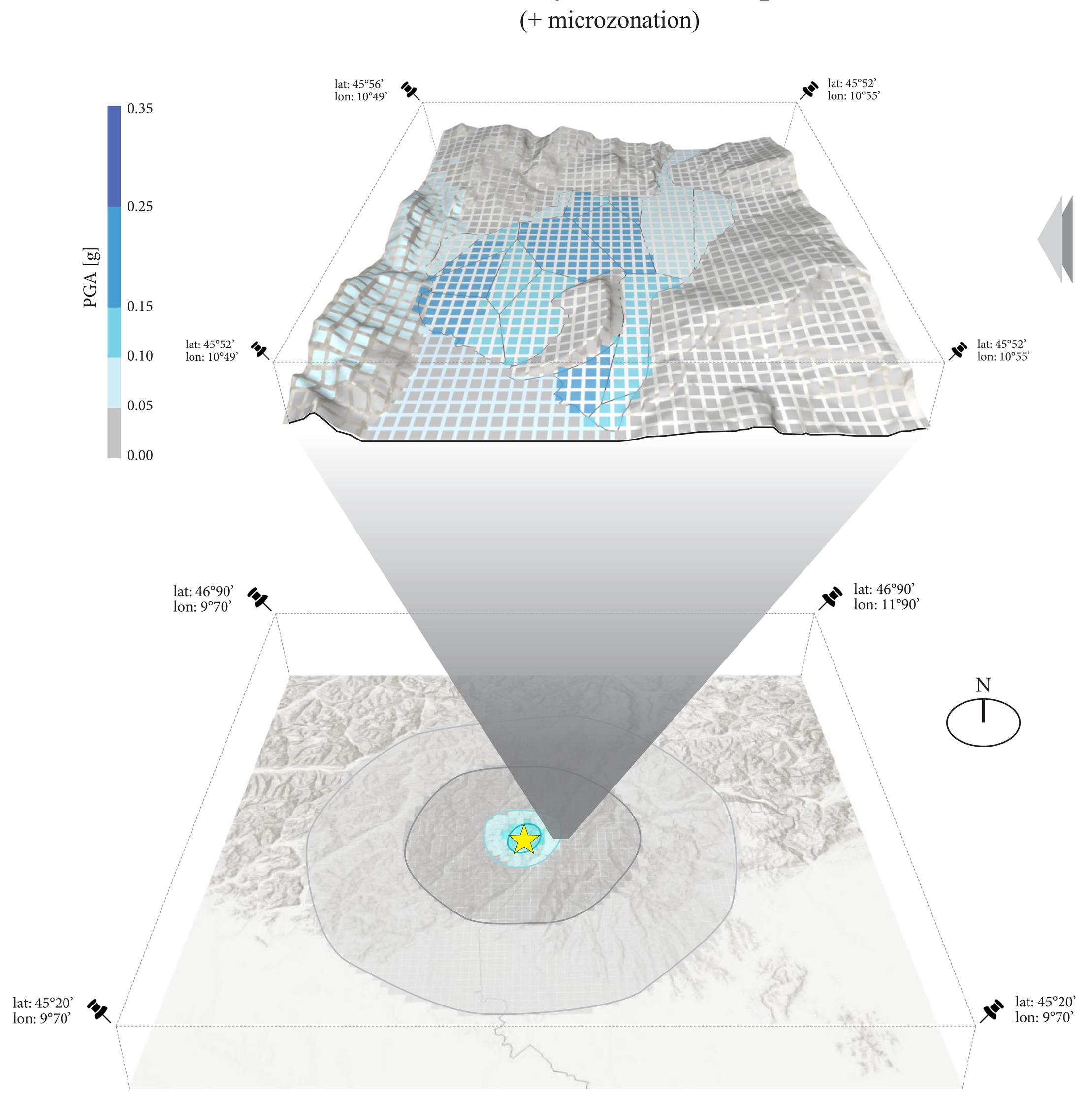


SEISMO-TECTONIC SETTINGS



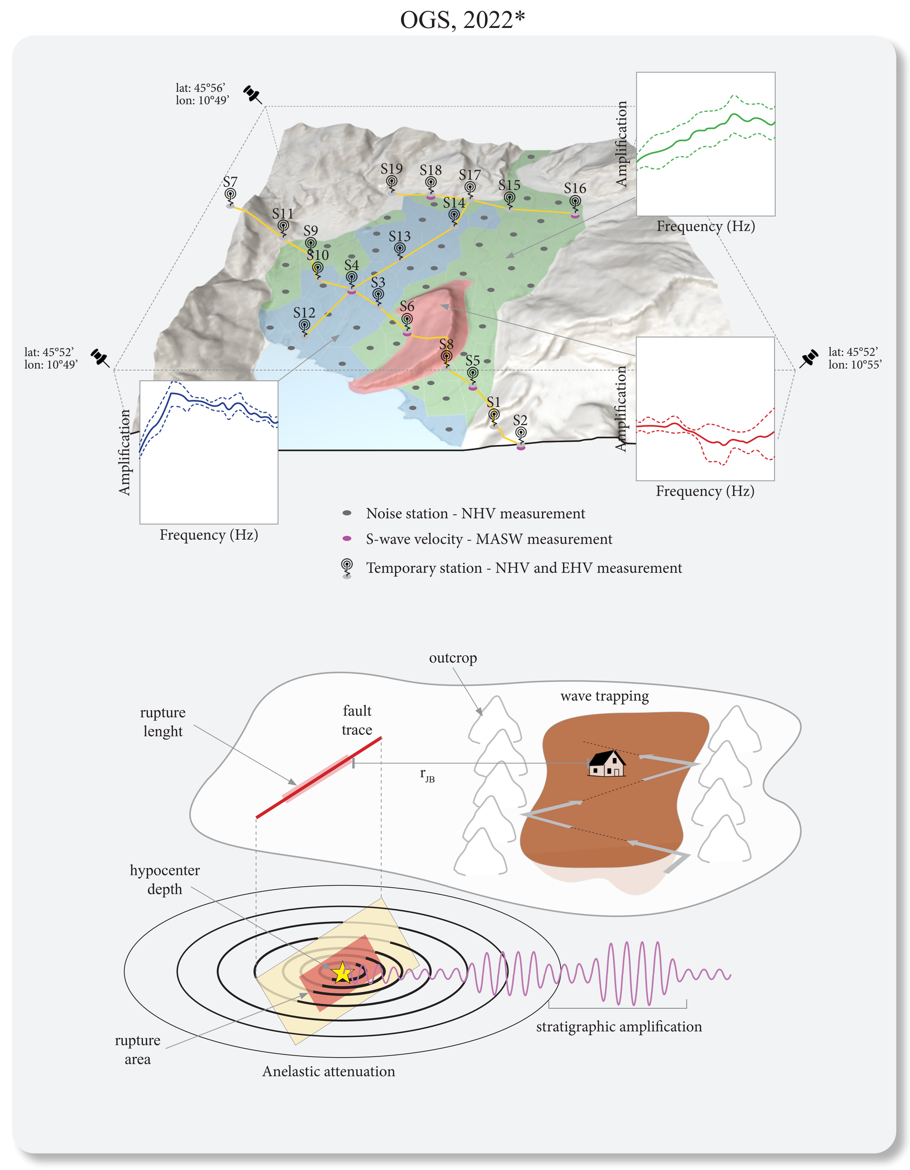
SCENARIO-BASED HAZARD MODELING

Intensity Measure map
(+ microzonation)



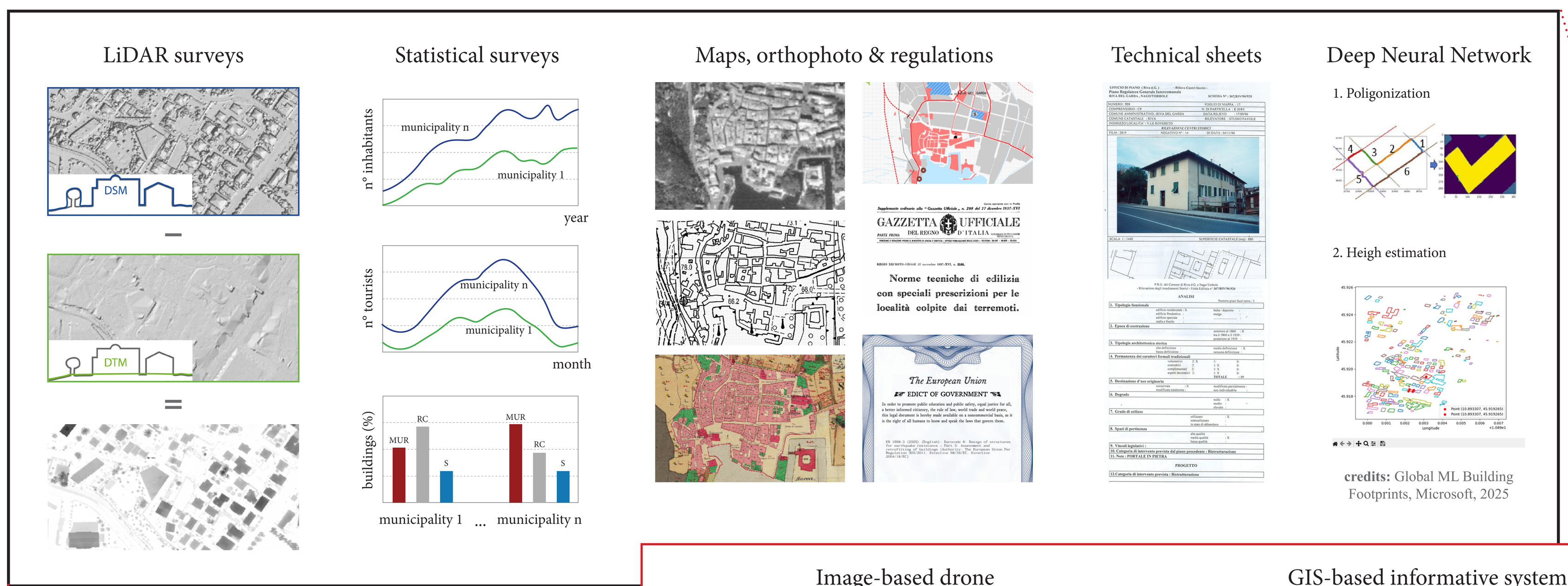
event	*date*	*time*	*hypocentral depth*	*latitude*	*longitude*	*local magnitude*	*focal mechanism*	*rupture area*	M_0
S.Lucia	13/12/76	06:24 am	6 km	45.9035 °N	10.7690 °E	4.3	strike slip	~3 km^2	3.16e15 Nm
	Molin, 1979		OGS, 2019	Viganò et al., 2021		Sandron et al., 2015	Saraò et al., 2020	Wells and Coppersmith, 1994	Kanamori and Anderson, 1975

Microzonation
OGS, 2022*



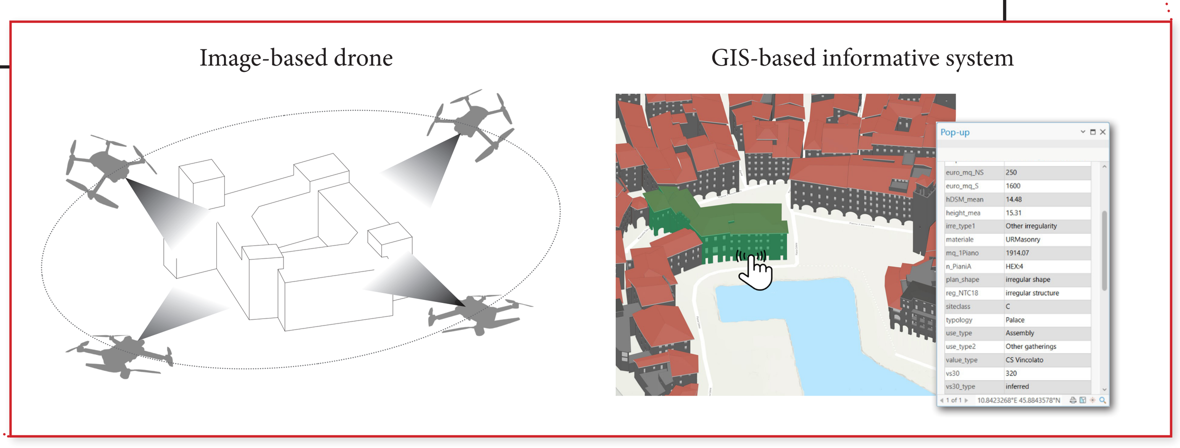
*credits: G. Laurenzano, 2022: "Studio riguardante la risposta sismica locale de tratto terminale della valle del fiume Sarca in prossimità del Lago di Garda"

EXPOSURE & FRAGILITY FRAMEWORK



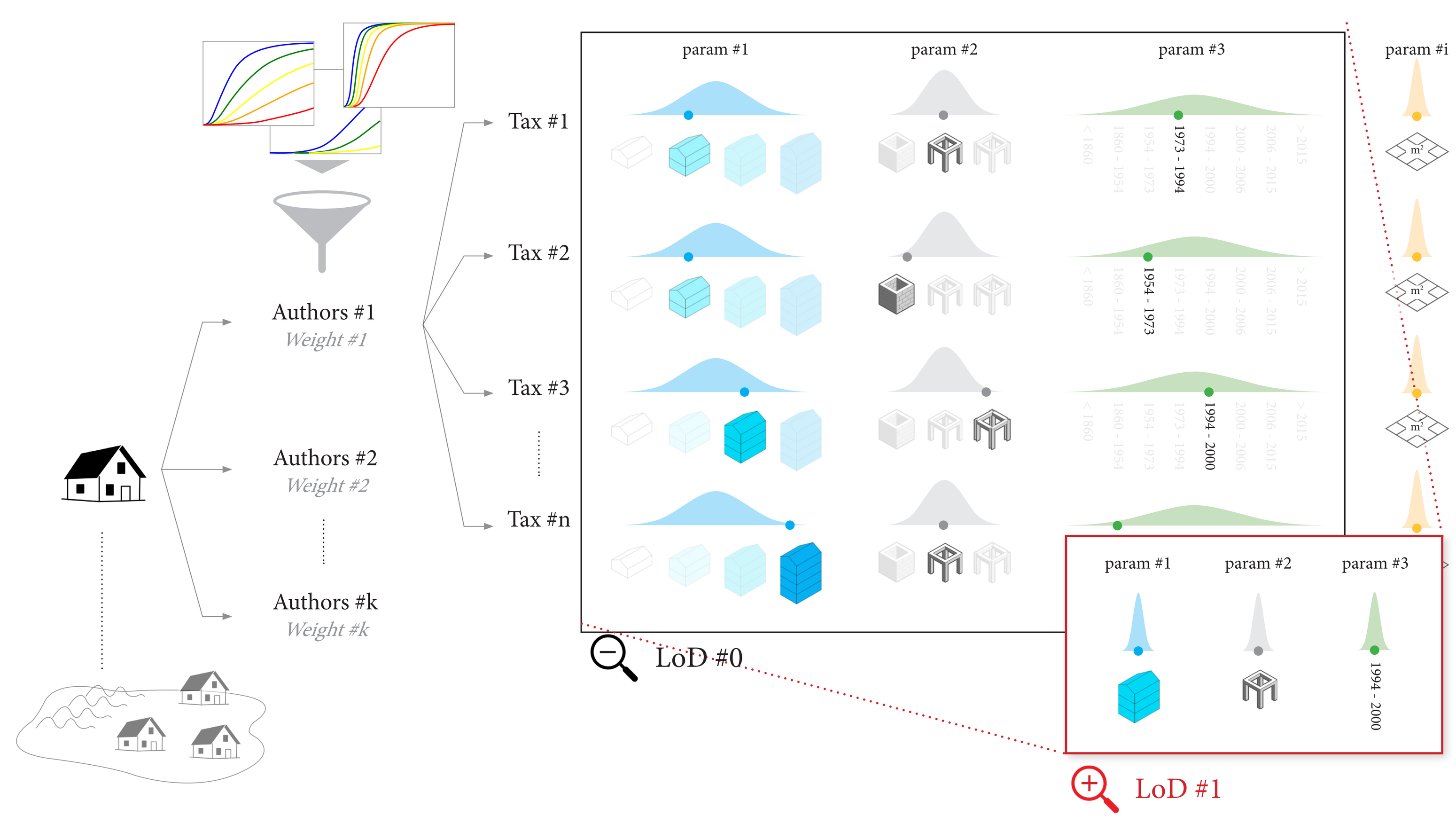
LoD #0

The LoD#0 is a coarse exposure model obtained from data fusion and harmonization process for the entire building stock; therefore, it contains uncertain or incomplete attributes. A refinement process (LoD#1) was conducted for a pilot area to reduce uncertainties and therefore obtain better damage and losses estimations.



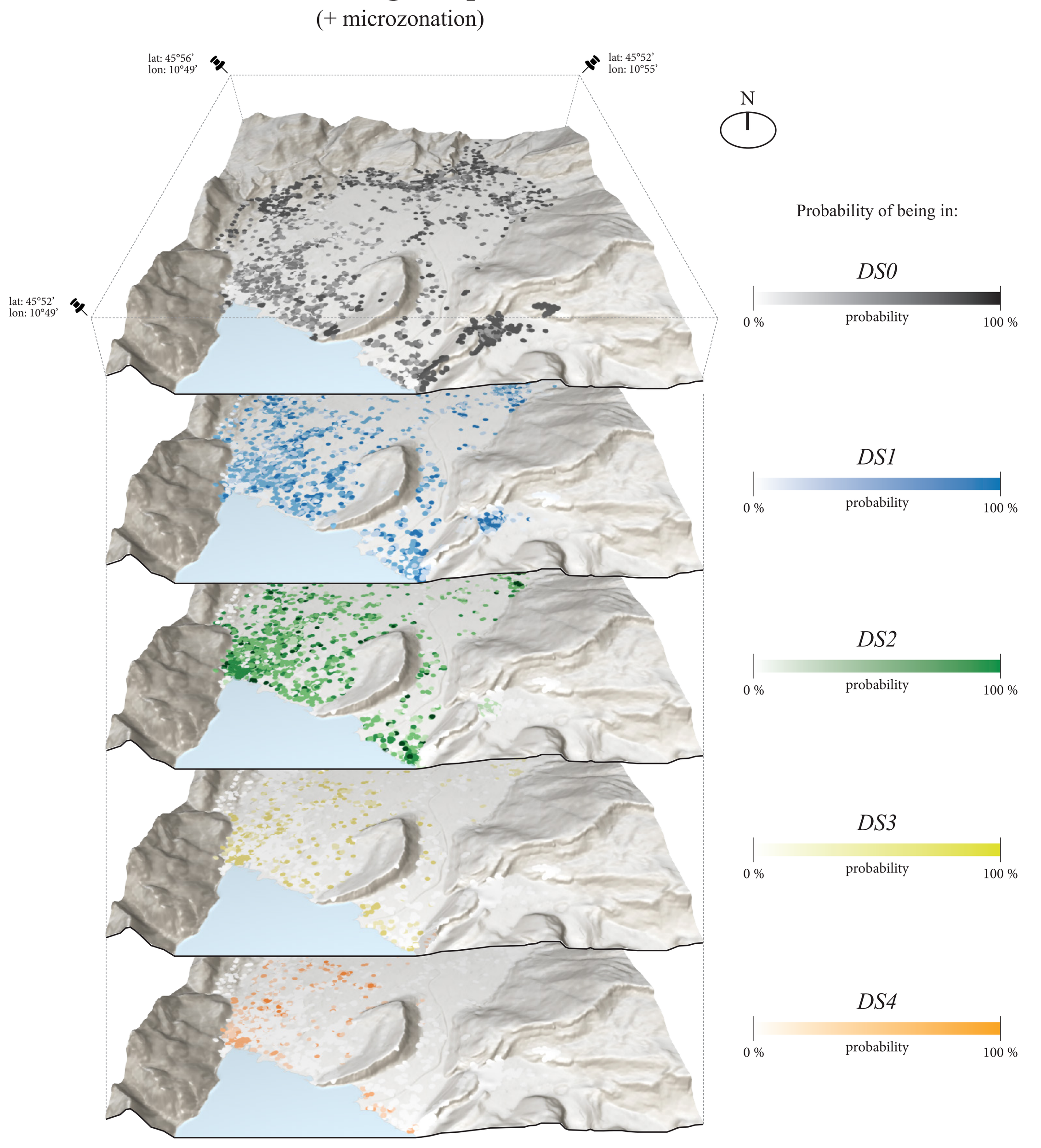
LoD #1

Attributes uncertainties at LoD#0, and suitable Euro-mediterranean fragility models retrieved from literature, are ensembled together into a logic tree approach. To account for exposure variability, multiple plausible taxonomies are generated through a Monte Carlo sampling informed by regional or censuns distributions. To account for fragility models variability, multiple authors where selected and weighted according to their representativeness with respect to the recurrent or dominant classess of the building stock under study.

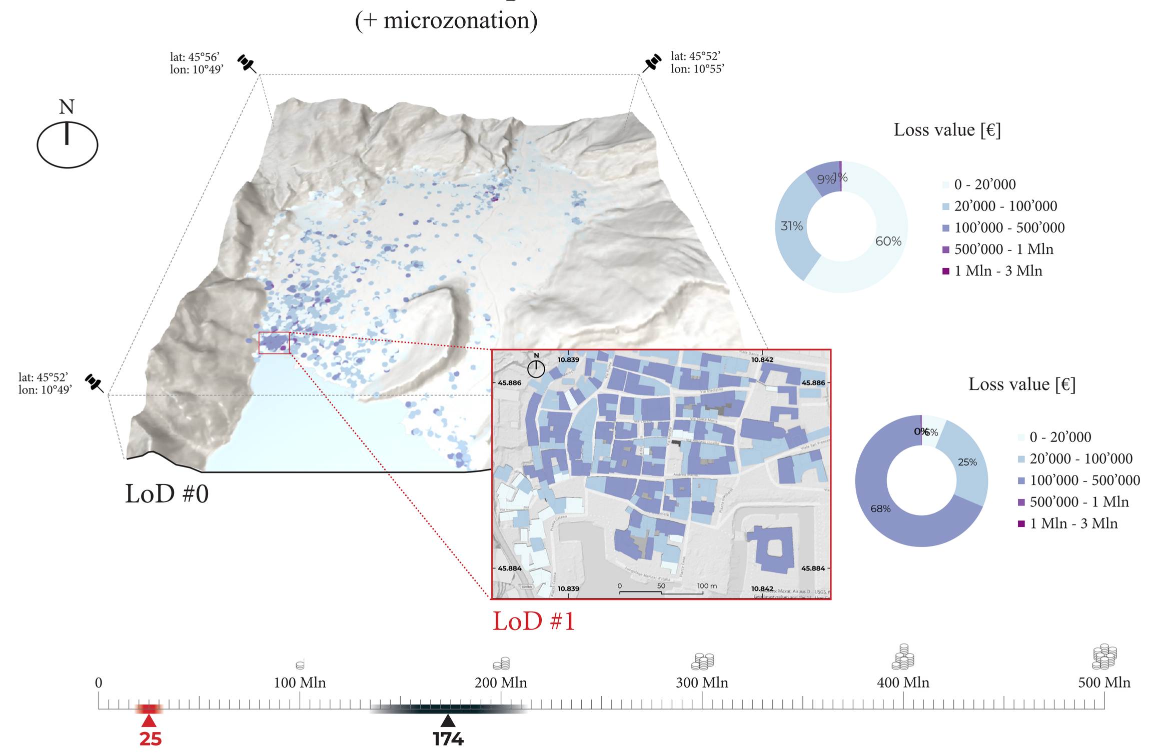


DAMAGE & LOSSES RESULTS

Damage maps
(+ microzonation)

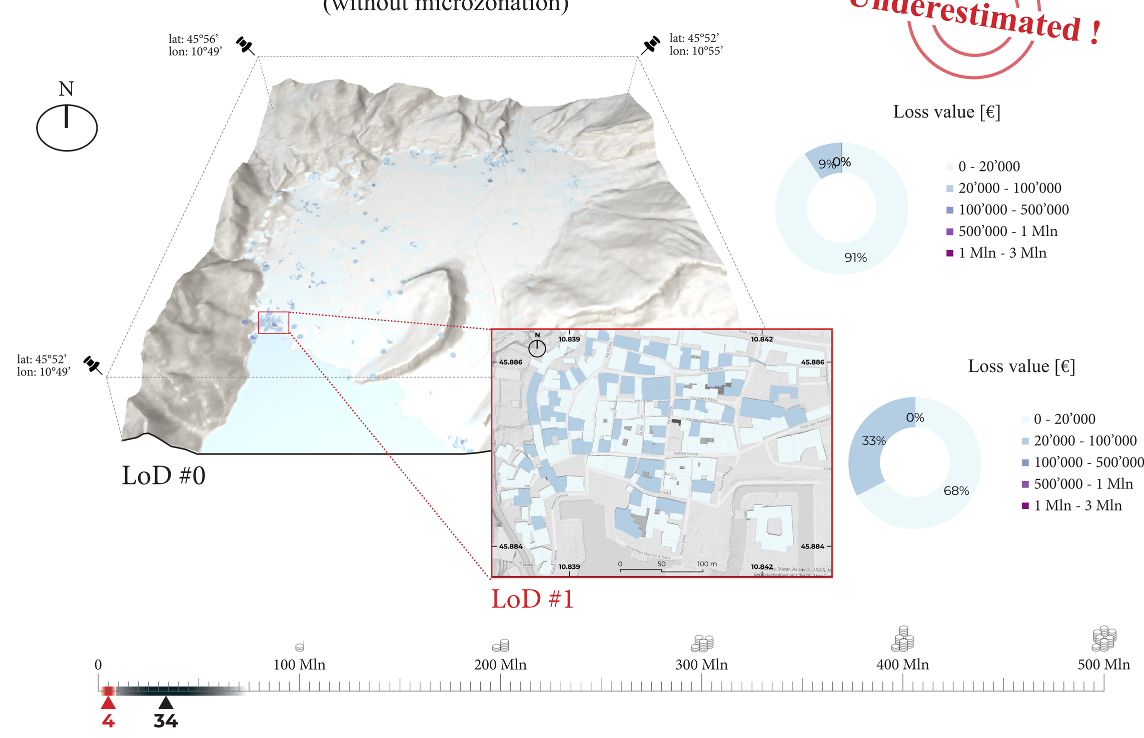


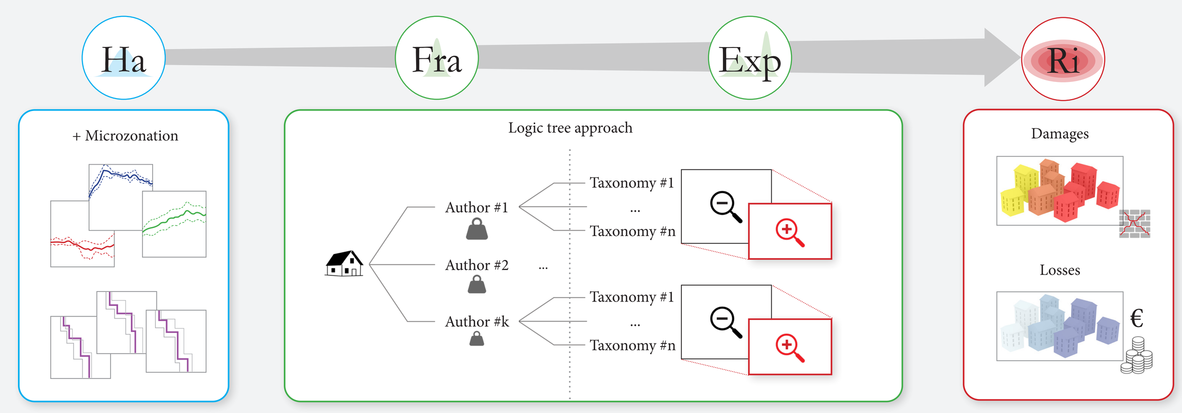
Losses maps
(+ microzonation)



Losses maps
(without microzonation)

Underestimated !





NEXT STEPS

- Represent lolcal site effects by site response analiysis
- Develop a site-specific GMPE for the basin area
- Develop a lolcal spatial-cross-correlation model
- Extend the risk assessment over all Trentino region
- S.Lucia earthquake with phyisics-based simulations

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scan me!

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